

WASP-43b

R-0585

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230124 · created 2026-07-24T20:15:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459968.8133 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.058
Transit depth (Rp/Rs)^2	3.91 +/- 2.29 [%]
Orbital Inclination (inc)	86.54 +/- 2.81
Airmass coefficient 1 (a1)	1.39 +/- 0.72
Airmass coefficient 2 (a2)	-0.17 +/- 0.22
Scatter in the residuals of the lightcurve fit is	52.11 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.061 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.058 vs 0.1594 (deviation 0.53σ)
Duration fit vs published	0.061 d vs 0.0512 d (deviation 1.59σ)
Mid-time O–C	4.5 min
Depth SNR	2.7
Residual scatter	52.11 %

Light curve

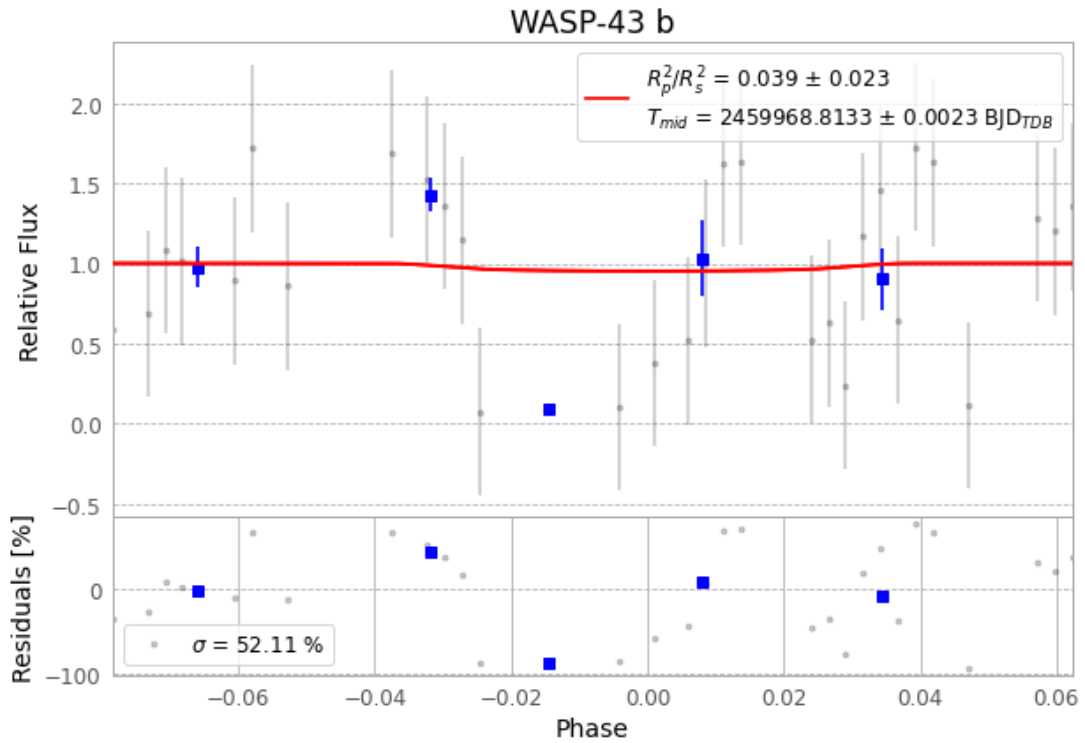


Figure 1 — FinalLightCurve_WASP-43 b_2023-01-23.png (SHA-256 6d8307b16e9e3e2b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [602, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0428820049c60692...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

63 FITS frames (41 MB), WASP-43230124054515.FITS → WASP-43230124085115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230124.log (c90ce93efda0...), FinalLightCurve_WASP-43 b_2023-01-23.png (6d8307b16e9e...), FinalLightCurve_WASP-43 b_2023-01-23.csv (309e8a9c2fbd...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 20:15Z.